

Hidayat Ur Rehman

C++ Software Engineer | Systems Software | Windows and Linux

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Professional Summary

Systems-focused C++ Software Engineer with experience developing cross-platform Windows and Linux solutions for data erasure, storage systems, financial trading, and automotive software. Strong in modern C++ (C++11/14/17/20), object-oriented design, memory management, concurrency, debugging, CMake, and test-driven development. Currently contributing to Blancco's core erasure products across HDD, SSD, NVMe, LUN, Unix, macOS, and removable-media workflows.

Technical Skills

Modern C++:

C++11/14/17/20, C++ STL, RAII, smart pointers, move semantics, templates, exception safety, object-oriented programming, design patterns

Windows and systems:

Windows, Linux, Unix, macOS, Win32, MFC, Qt, MSI, Linux system programming, file systems, storage systems, HDD, SSD, NVMe

Build and debugging:

CMake, Visual Studio, GCC, Git, GitHub, SVN, Jenkins, GDB, Visual Studio Debugger, CI/CD

Concurrency and quality:

Multithreading, mutexes, condition variables, atomics, thread safety, IPC, socket programming, memory management, code optimization, code reviews

Testing:

Unit testing, integration testing, Google Test, test-driven development, production debugging

AI-assisted development:

GitHub Copilot, Claude Console

Additional:

Python, SQL, Bash, Boost, Docker, REST APIs, JSON, XML, TCP/IP, SSL/TLS, PostgreSQL, MySQL, InfluxDB, FIX Protocol

Professional Experience

Software Engineer

Feb 2025 -- Present

Blancco Oy Ltd., Finland [Onsite]

- Develop and maintain core C++ software for Blancco PreInstall, Blancco File Erasure, LUN Erasure, LUN for Unix, LUN for macOS, and Removable Media Erasure.
- Optimize low-level storage access and secure data-sanitization workflows for HDD, SSD, and NVMe devices while supporting NIST and DoD compliance requirements.
- Revamped the Removable Media Erasure user interface using C++, MFC, and Windows desktop development, improving usability and maintainability.
- Built a PreInstall MSI generator using Qt and Python to automate installer generation and improve deployment reliability.
- Refactor legacy C++ with modern C++17/C++20 practices and troubleshoot concurrency, memory-management, and storage-related defects across Windows and Linux.
- Contribute to architecture discussions, code reviews, debugging, testing, and continuous improvement across multiple production C++ codebases.
- Reduced erasure time by 25%, certification failures by 40%, platform-specific defects by 35%, and critical runtime errors by 50% through targeted engineering improvements.

Software Engineer

Apr 2024 -- Dec 2024

Logiciel Services LLC, USA [Remote]

- Optimized NYSE order delivery using modern C++, reducing latency by 20% and increasing throughput by 25%.
- Implemented FIX Protocol messaging for financial trading, reducing message-processing time by 30%.
- Developed market-data services with PostgreSQL and migrated InfluxDB from v1 to v2, improving query performance by 35%.

Senior Software Engineer

Dec 2022 -- Apr 2024

Avanza Solutions Pvt. Ltd. [Hybrid]

- Developed and optimized C++ software, debugged production defects, and provided technical support to banking customers.
- Served as resident engineer at U Microfinance Bank and deployed a RAAST payment-system module.

Development Engineer, C++

Mar 2022 -- Nov 2022

Sedenius Engineering, Germany [Hybrid]

- Developed high-performance C++ automotive software for CARIAD, applying memory-management, optimization, code-review, and mentoring practices.

Research Officer

Sep 2021 -- Apr 2022

Shaheen Foundation, Pakistan [Onsite]

- Developed technical solutions across research and software projects.

Education

Master of Computer Science and Engineering

2024 -- 2026

University of Oulu, Finland [Graduated]

Master of Cyber Security

2021 -- 2023

Air University, Islamabad, Pakistan [CGPA: 3.57/4]

Bachelor of Computer Science

2017 -- 2021

Air University, Islamabad, Pakistan [CGPA: 3.01/4]

Selected Projects

Master's Thesis: Wearable IMU Hand Rehabilitation with TinyML

[Raspberry Pi Pico WH, ICM-20948, Python, 1D-CNN | 2026]

Developed a three-IMU wearable system for classifying six hand rehabilitation movements: Flexion, Extension, Pronation, Supination, Radial Deviation, and Ulnar Deviation. Connected ICM-20948 sensors to a Raspberry Pi Pico WH through I2C, collected movement data at 100 Hz, and evaluated SVM, KNN, Random Forest, 1D-CNN, and BiLSTM models.

- Achieved 98.31% accuracy with three sensors compared with 95.36% using one sensor; SVM reached 99.16% test accuracy and 1D-CNN achieved 99.48% mean cross-validation accuracy.
- Selected and deployed a fully quantized 1D-CNN model with TFLite Micro for on-device TinyML inference on a resource-limited embedded system.

APT Detection Tool

[Python, machine learning, Flask, network traffic | 2024]

Trained machine and deep learning models for advanced persistent threat detection, captured HTTP, TCP, and UDP traffic, and deployed extracted features to a Flask-based model on a local network.

Removable Media and USB Monitoring Tool

[C++, Python, SQL, socket programming | 2022]

Built a monitoring tool that detects connected USB devices, logs device details, and provides real-time alerts for unauthorized activity.

Dark Web Crawler -- Bachelor's Final Project

[Python, Flask, MongoDB, Elasticsearch, Socket.IO, Tor | 2021]

Built a crawler that discovers and collects internal URLs from onion sites, writes discovered links to `onionlinks.txt`, and uses queue-based crawling with multithreading and multiprocessing for efficient collection.